Question - 17: Write a Prolog program to find the sum of integers from 1 to N.

Program Code

```
sum_n(0, 0).

sum_n(N, Sum) :-
    N > 0,
    N1 is N - 1,
    sum_n(N1, PreviousSum),
    Sum is N + PreviousSum.
```

Program Output

```
PS E:\Desktop\SIMATS\Artificial Intelligence\AI LAB> swipl --version
SWI-Prolog version 10.0.2 for x64-win64
PS E:\Desktop\SIMATS\Artificial Intelligence\AI LAB> swipl -s experiment17_
m_integers.pl
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

1 ?- sum_n(5, Sum).
Sum = 15
```